



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Ambulance and Dispatch System

Team ID: DSCPP-III-2026-T148

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T148

Semester: 3rd

Section: K, E, ML1, ML3

Domain: *B.tech CSE*

Team Members

1. *Neel Hajariwala – 2510011575*
2. *Keshav Namdev – 2510360097*
3. *Lichhavi Chauhan – 2510370231*
4. *Aditya Ahlawat – 251251000*

Mentor

Mr.Ramesh Singh Rawat

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

In emergencies, manual ambulance dispatch can cause delays in prioritizing cases and assigning the nearest available ambulance. This project addresses this gap by developing a C++ and Data Structures-based system that efficiently prioritizes emergency requests and dispatches suitable ambulances, improving response time and resource utilization.

Motivation

- ▶ *Delays*
- ▶ *Inefficacy*
- ▶ *Mortality*

Target Users / Stakeholders

- ▶ *Dispatchers*
- ▶ *Hospitals*
- ▶ *Efficiency*

Objectives

Project Objectives

1. *Develop: C++ dispatch system.*
2. *Implement: Priority queues and graph algorithms.*
3. *Integrate: DSA and OOP principles.*
4. *Evaluate: Response times and queue speeds.*
5. *Demonstrate: Scalable real-time ambulance routing.*

Key Objective: *Minimize dispatch delay via automated priority routing.*

Proposed Solution

Proposed Approach

1. *Input: Call data.*
2. *Processing: Priority routing.*
3. *Logic: Auto allocation.*
4. *Database: Status logs.*
5. *Output: Dispatch details.*

Key Features

- ▶ *Triage: Max-Heap queues.*
- ▶ *Routing: Dijkstra paths.*
- ▶ *Tracking: Live monitoring.*
- ▶ *Speed: Zero delays.*
- ▶ *Impact: Faster care.*

Automates patient triage using Max-Heap priority queues and optimizes ambulance navigation via Dijkstra's shortest-path algorithm, directly eliminating manual dispatch delays and minimizing emergency response times.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Linked Lists, Trees, Graphs, Hashing, etc.</i>	<i>Triage Engine</i>
OOPs with C++	<i>Classes, Inheritance, Polymorphism, STL, etc.</i>	<i>Dispatch Manager</i>
Operating Systems	<i>Processes, Scheduling, Memory, Synchronization, etc.</i>	<i>Request Queue</i>
DBMS	<i>SQL, Transactions, Normalization, Indexing, etc.</i>	<i>Fleet Database</i>

Requirement: Integrates dynamic C++ data structures with automated system architecture.

System Architecture / Workflow



Major Components

- ▶ *Triage Engine*
- ▶ *Routing Module*
- ▶ *Database Layer*

Data / Control Flow

- ▶ *Sequential*
- ▶ *Automated*
- ▶ *Assignment*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++*
- ▶ Tools / Framework: *STL*
- ▶ IDE: *VS Code*
- ▶ Version Control: *GitHub*

Development Methodology

1. Emergency Triage Scoping.
2. Heap logic.
3. C++ Core Coding.
4. Dijkstra Path Validation.
5. Performance optimization.

C++ delivers high-performance execution for instant pathfinding, file-based logging ensures lightweight zero-overhead storage, and Agile methodology enables rapid, iterative testing of critical emergency logic.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Scoping*
- ▶ *Design*
- ▶ *Prototyping*
- ▶ *Structuring*
- ▶ *Datasets*

Individual Contribution

Member	Role	Contribution
M1	Lead	Neel Hajariwala
M2	Developer	Keshav Namdev
M3	Developer/Docs	Lichhavi Chauhan
M4	DB/Testing	Aditya Ahlawat

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Architecture & Triage Design
2. Phase-II: Graph Routing & Heap Coding
3. Phase-III: System Testing & Integration

Expected Outcomes

- ▶ *Working C++ dispatch prototype*
- ▶ *Reduced emergency response times*
- ▶ *Functional shortest-path routing*
- ▶ *Scalable multi-ambulance queue*

References

1. *Journal - Introduction to Algorithms*
2. *Book - Stroustrup*
3. *Map and Road Network Data*
4. *OSRM - Open Source Routing Machine*

Thank You

Questions & Suggestions